A GUIDE FOR STUDENTS

Critical Thinking Toolkit

How to question ideas, weigh evidence and think for yourself

Critical thinking is the skill Oxbridge tutors test more than any other. It runs through admissions tests, interviews and every tutorial or supervision you'll have at university.

Despite its name, critical thinking isn't about being negative or finding fault with everything. It's about thinking clearly and carefully: understanding ideas properly, weighing them fairly and reaching your own well-reasoned conclusions. This toolkit gives you practical techniques you can start using straight away.

SECTION ONE

What critical thinking really means

Asking good questions

Critical thinkers don't accept ideas at face value. They ask why something is true, how we know, and what follows from it.

Understanding before judging

Before deciding whether an argument is convincing, make sure you understand exactly what it's saying and why.

Separating evidence from opinion

A strong view isn't the same as a well-supported one. Critical thinkers look for the evidence behind a claim.

Being willing to change your mind

Good thinkers update their views when they meet better arguments or new evidence. This is a strength, not a weakness.

> **WORTH KNOWING**
> Critical thinking isn't about tearing ideas down. It's about understanding why something is or isn't convincing, and that includes recognising when an argument is strong.

SECTION TWO

Breaking down an argument

Most arguments are made up of the same basic parts. Learning to spot them makes any argument easier to understand and assess.

Claim or conclusion

The main point the argument is trying to prove.

Reasons

The points offered to support the claim.

Evidence

The facts, data or examples that back up the reasons.

Assumptions

Ideas the argument takes for granted without stating them. These are often where weaknesses hide.

AN EXAMPLE

"Students who read for pleasure tend to get higher grades. Therefore, schools should require every student to read a novel each week."

Conclusion: Schools should require every student to read a novel each week.

Reason and evidence: Students who read for pleasure tend to get higher grades.

Hidden assumptions: That reading causes the higher grades, rather than both being linked to something else, and that reading because you're required to has the same effect as reading for pleasure. Both are open to question.

SECTION THREE

Your toolkit

1 Spot the assumption

Ask what the argument takes for granted. Unstated assumptions are often the weakest link, and questioning them can change how convincing an argument seems.

TRY THIS

- For any argument you read, write down one assumption it relies on.
- Ask whether that assumption is reasonable.

2 Question the evidence

Consider whether the evidence is reliable, relevant and enough to support the conclusion. Where does it come from, and could it be biased or incomplete?

TRY THIS

- Ask: who produced this evidence, and how?
- Check whether the evidence actually supports the specific claim being made.

3 Consider alternatives

Before accepting an explanation, ask whether something else could explain the same facts. Good thinkers consider several possibilities.

TRY THIS

- Come up with at least one alternative explanation for any claim.
- Think about what evidence would help you decide between them.

4 Check the logic

Even if every reason is true, the conclusion may not follow from them. Check whether there's a clear, logical link between the reasons and the conclusion.

TRY THIS

- Ask: if the reasons were true, would the conclusion have to be true?
- Look for any leap in reasoning that isn't explained.

5 Look for what's missing

Notice what an argument leaves out: important counterarguments, inconvenient evidence or groups of people it doesn't consider.

TRY THIS

- Ask what someone who disagreed would point out.
- Think about who or what the argument doesn't mention.

6 Test it with examples

Try applying an argument to a specific case. If it leads to a strange or unacceptable result, there may be a problem with it.

TRY THIS

- Think of a real or imagined example the argument should cover.
- Ask whether the argument still works in that case.

SECTION FOUR

Common flaws in reasoning

Confusing correlation with causation

Just because two things happen together doesn't mean one causes the other. Something else may explain both.

Generalising from too few examples

A handful of cases can't prove a general rule. Ask whether the examples are typical.

Attacking the person, not the argument

Criticising who is making an argument doesn't show the argument itself is wrong.

Presenting a false choice

Arguments that offer only two options often ignore other possibilities.

Appealing to popularity

An idea isn't true just because many people believe it.

SECTION FIVE

Practising critical thinking

Question the news

When you read or watch a news story, identify its main claim and the evidence behind it.

Debate with others

Discussing ideas with friends and family, especially when you disagree, sharpens your reasoning.

Analyse your reading

Use the tools in this guide on the books and articles you read for your subject.

Practise with your mentor

Your Access Oxbridge mentor can challenge your thinking, test your reasoning with unfamiliar questions and help you prepare for the kind of discussion you'll meet at interview.

Critical thinking checklist

- [ ] I understand what the argument is claiming
- [ ] I've identified its reasons and evidence
- [ ] I've spotted its assumptions
- [ ] I've questioned the evidence
- [ ] I've considered alternative explanations
- [ ] I've checked the conclusion follows
- [ ] I've looked for what's missing
- [ ] I've reached my own judgement

Where to find official information

Details such as admissions tests, deadlines and course requirements can change from year to year. Always check the latest information on the official websites:

- **University of Oxford undergraduate admissions:** ox.ac.uk/admissions/undergraduate
- **University of Cambridge undergraduate study:** undergraduate.study.cam.ac.uk
- **UCAS:** ucas.com